

CHRIST

(DEEMED TO BE UNIVERSITY)

BANGALORE | DELHI NCR | PUNE

Regression Analysis Lab

Course Code: MST504-A

JOINS

LAB: 3

Submitted by

RAGHURAAM P

2648130

MSc Statistics

Submitted to

Dr. Monisha Singh

Department of Statistics and Data Science

Creation of table

```sql
CREATE DATABASE lab_3;
USE lab_3;
CREATE TABLE employee
(empid INT PRIMARY KEY,
name VARCHAR(50),
deptid INT,
salary INT,
city VARCHAR(50)
);
INSERT INTO employee VALUES
(1, 'Arun', 101, 60000, 'Bangalore'),
(2, 'Rahul', 102, 45000, 'Chennai'),
(3, 'Priya', 103, 70000, 'Bangalore'),
(4, 'Sneha', 101, 55000, 'Mumbai'),
(5, 'Kiran', 104, 40000, 'Chennai'),
(6, 'Divya', NULL, 35000, 'Bangalore');

CREATE TABLE Department
(
    deptid INT PRIMARY KEY,
    deptname VARCHAR(50),
    location VARCHAR(50)
);
INSERT INTO Department VALUES
(101, 'IT', 'Bangalore'),
(102, 'HR', 'Chennai'),
(103, 'Finance', 'Mumbai'),
(104, 'Sales', 'Delhi'),
(105, 'Marketing', 'Hyderabad');
```

QUESTIONS CODES WITH OUTPUT

1. Display employee names along with their department names using INNER JOIN.

CODE:

```
SELECT employee.name , Department.deptname
FROM employee INNER JOIN Department
ON employee.deptid = Department.deptid;
```

OUTPUT :

	name	deptname
▶	Arun	IT
	Rahul	HR
	Priya	Finance
	Sneha	IT
	Kiran	Sales

2. Display employee ID, employee name and department ID from both tables.

CODE:

```
SELECT employee.empid , employee.name , Department.deptid
FROM employee JOIN Department
ON employee.deptid = Department.deptid;
```

OUTPUT:

	empid	name	deptid
▶	1	Arun	101
	2	Rahul	102
	3	Priya	103
	4	Sneha	101
	5	Kiran	104

3. Display employee names and department locations.

CODE:

```
SELECT employee.name , Department.location
FROM employee JOIN Department
ON employee.deptid = Department.deptid;
```

OUTPUT:

	name	location
▶	Arun	Bangalore
	Rahul	Chennai
	Priya	Mumbai
	Sneha	Bangalore
	Kiran	Delhi

4. Display employees working in the IT department.
   CODE:

```
SELECT * FROM employee JOIN Department
ON employee.deptid = Department.deptid
WHERE Department.deptname = 'IT';
```

   OUTPUT:

	empid	name	deptid	salary	city	deptid	deptname	location
▶	1	Arun	101	60000	Bangalore	101	IT	Bangalore
	4	Sneha	101	55000	Mumbai	101	IT	Bangalore

Bangalore

5. Display employees working in the HR department.
   CODE:

```
SELECT * FROM employee JOIN Department
ON employee.deptid = Department.deptid
WHERE Department.deptname = 'HR';
```

   OUTPUT:

	empid	name	deptid	salary	city	deptid	deptname	location
▶	2	Rahul	102	45000	Chennai	102	HR	Chennai

6. Display employee names, salaries and department names.
   CODE:

```
SELECT employee.name, employee.salary, Department.deptname
FROM employee JOIN Department
ON employee.deptid = Department.deptid;
```

   OUTPUT:

	name	salary	deptname
▶	Arun	60000	IT
	Rahul	45000	HR
	Priya	70000	Finance
	Sneha	55000	IT
	Kiran	40000	Sales

7. Display employees who belong to the Finance department.
   CODE:

```
SELECT * FROM employee JOIN Department
ON employee.deptid = Department.deptid
WHERE Department.deptname = 'Finance';
```

   OUTPUT:

	empid	name	deptid	salary	city	deptid	deptname	location
▶	3	Priya	103	70000	Bangalore	103	Finance	Mumbai

8. Display all employee names and department names using LEFT JOIN.
   CODE:

```
SELECT employee.name , Department.deptname
FROM employee LEFT JOIN department
ON employee.deptid = Department.deptid;
```

   OUTPUT:

	name	deptname
▶	Arun	IT
	Rahul	HR
	Priya	Finance
	Sneha	IT
	Kiran	Sales
	Divya	NULL

9. Display all department names and employee names using RIGHT JOIN.
   CODE:

```
SELECT employee.name , Department.deptname
FROM employee RIGHT JOIN department
ON employee.deptid = Department.deptid;
```

   OUTPUT:

Result Grid | Filter Row

name	deptname
Sneha	IT
Arun	IT
Rahul	HR
Priya	Finance
Kiran	Sales
NULL	Marketing

10. Display employees working in Bangalore along with their department names.
    CODE:

```
SELECT * FROM employee JOIN Department
ON employee.deptid = Department.deptid
WHERE employee.city = 'Bangalore';
```

    OUTPUT:

empid	name	deptid	salary	city	deptid	deptname	location
1	Arun	101	60000	Bangalore	101	IT	Bangalore
3	Priya	103	70000	Bangalore	103	Finance	Mumbai

11. Display employees whose salary is greater than ₹50,000 along with their department names.
    CODE:

```
SELECT employee.name , employee.salary , Department.deptname
FROM employee JOIN Department
ON employee.deptid = Department.deptid
WHERE employee.salary > 50000;
```

    OUTPUT:

name	salary	deptname
Arun	60000	IT
Priya	70000	Finance
Sneha	55000	IT

12. Display employees who belong to either IT or HR departments.
    CODE:

```
SELECT * FROM employee JOIN Department
ON employee.deptid = Department.deptid
WHERE Department.deptname IN ('IT','HR');
```

    OUTPUT:

empid	name	deptid	salary	city	deptid	deptname	location
1	Arun	101	60000	Bangalore	101	IT	Bangalore
2	Rahul	102	45000	Chennai	102	HR	Chennai
4	Sneha	101	55000	Mumbai	101	IT	Bangalore

13. Display all employees, including those who are not assigned to any department.
CODE:

```
SELECT *
FROM employee
LEFT JOIN Department
ON employee.deptid = Department.deptid;
```

OUTPUT:

empid	name	deptid	salary	city	deptid	deptname	location
1	Arun	101	60000	Bangalore	101	IT	Bangalore
2	Rahul	102	45000	Chennai	102	HR	Chennai
3	Priya	103	70000	Bangalore	103	Finance	Mumbai
4	Sneha	101	55000	Mumbai	101	IT	Bangalore
5	Kiran	104	40000	Chennai	104	Sales	Delhi
6	Divya	106	35000	Bangalore	NULL	NULL	NULL

14. Display departments that do not have any employees.
CODE:

```
SELECT Department.deptname
FROM employee
RIGHT JOIN Department
ON Department.deptid = employee.deptid
WHERE employee.empid IS NULL;
```

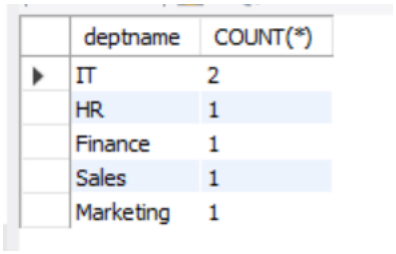
OUTPUT:

deptname
Marketing

15. Display the total number of employees in each department.
CODE:

```
SELECT Department.deptname, COUNT(*)
FROM employee
RIGHT JOIN Department
ON Department.deptid = employee.deptid
GROUP BY Department.deptname;
```

OUTPUT:



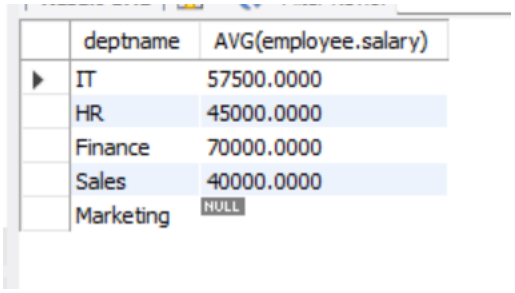
deptname	COUNT(*)
IT	2
HR	1
Finance	1
Sales	1
Marketing	1

16. Display the average salary of employees in each department.

CODE:

```
SELECT Department.deptname, AVG(employee.salary)
FROM employee
RIGHT JOIN Department
ON Department.deptid = employee.deptid
GROUP BY Department.deptname;
```

OUTPUT:



deptname	AVG(employee.salary)
IT	57500.0000
HR	45000.0000
Finance	70000.0000
Sales	40000.0000
Marketing	NULL

17. Display the maximum salary in each department.

CODE:

```
SELECT Department.deptname, MAX(employee.salary)
FROM employee
RIGHT JOIN Department
ON Department.deptid = employee.deptid
GROUP BY Department.deptname;
```

OUTPUT:

deptname	MAX(employee.salary)
IT	60000
HR	45000
Finance	70000
Sales	40000
Marketing	NULL

18. Display departments having more than one employee.
    CODE:

```
SELECT Department.deptname
FROM employee
RIGHT JOIN Department
ON Department.deptid = employee.deptid
GROUP BY Department.deptname
HAVING COUNT(employee.name)>1;
```

    OUTPUT:

deptname
IT

19. Display employees whose department location is different from their city.
    CODE:

```
SELECT employee.name ,employee.city, Department.location
FROM employee JOIN Department
ON Department.deptid = employee.deptid
WHERE employee.city != Department.location;
```

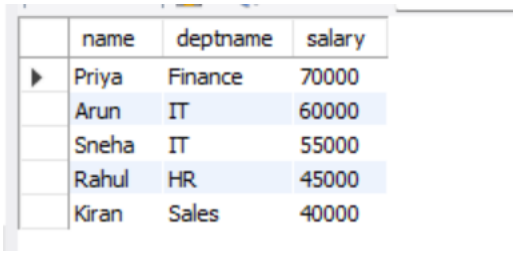
    OUTPUT:

name	city	location
Priya	Bangalore	Mumbai
Sneha	Mumbai	Bangalore
Kiran	Chennai	Delhi

20. Display employee names and department names in descending order of salary.
    CODE:

```
SELECT employee.name , Department.deptname, employee.salary
FROM employee JOIN Department
ON Department.deptid = employee.deptid
ORDER BY employee.salary DESC;
```

OUTPUT:



name	deptname	salary
Priya	Finance	70000
Arun	IT	60000
Sneha	IT	55000
Rahul	HR	45000
Kiran	Sales	40000